

SIH26055 — Smart Scan Strategy for Electronic Warfare

1. Project Evidence

Evidence	Description	Purpose / Relevance
PS 26055	DRDO Smart India Hackathon 2026 problem statement: an ES/EW receiver has limited instantaneous bandwidth and must decide which narrow frequency slice to scan next with zero prior emitter intelligence.	Defines the core problem, operating constraints, and success measures such as detection/intercept latency, probability of intercept, and wasted dwells.
US10523342B1	ON! Systems patent describing reinforcement-learning-based receiver scan-schedule updates, including an actor-critic architecture and reward-based scheduling.	Primary prior-art reference showing that generic learning-driven scan scheduling is already claimed.
US11747438B1	BAE Systems patent for cognitive EW scheduling that assigns scan priority using threat levels for active emitters.	Second prior-art reference showing threat-priority scheduling in an EW setting.
SIH Differentiation	The problem statement explicitly combines zero prior emitter intelligence with frequency-hopping, scanning, deceptive decoy emitters, and reactive jamming.	Identifies the technical gap that the proposed SIH solution should target rather than claiming generic cognitive scheduling as new.
Proposed Method	Restless multi-armed bandit / POMDP belief-state framing, using an index policy such as Whittle index or a decaying-UCB variant, plus a decoy/deception filter.	Provides a mathematically distinct way to select the next band while balancing recent activity, uncertainty, and deception.
Evaluation	Benchmark against fixed-sweep and threat-priority baselines using detection latency, cumulative regret, and dwells wasted on decoys, including agile/frequency-hopping emitters.	Tests the specific uncovered difficulty rather than only demonstrating generic adaptive scanning.

2. Evidence Summary

Area	SIH26055 Evidence Summary
Core problem	An EW/ESM receiver can observe only a narrow slice of a wide RF spectrum. It must choose, dwell by dwell, what to scan next without prior knowledge of emitter frequencies, periods, or behaviour.
Threat environment	The stated environment includes frequency-hopping emitters, periodic scanning radars, deceptive decoy signals, and reactive jamming; the receiver therefore faces non-stationary and potentially adversarial observations.
Prior-art overlap	Both cited patents cover the broad concept of cognitively re-prioritising which band to scan. One uses reinforcement-learning schedule updates; the other uses threat-level priority for known/characterised emitters.
Technical gap	The report identifies the combination of zero prior emitter intelligence plus active frequency agility and deception as the central gap not clearly claimed by either cited patent.
Solution direction	Represent each frequency band with a changing belief about current emitter activity, then select the next band using a bandit/POMDP index policy. Penalise suspicious decoy-like patterns so the scheduler does not repeatedly follow a lure.
Success measures	Detection/intercept latency, probability of intercept, cumulative regret under non-stationarity, and dwells wasted on decoys. Results should include explicit comparisons with fixed-sweep and threat-priority baselines.
IP / research note	The source report recommends explicitly citing the two patents and differentiating the project technically. It also notes that any future fielding or commercialisation should receive a formal freedom-to-operate review.

Source basis: SIH26055 Prior-Art Comparison Report, including PS 26055, US10523342B1, US11747438B1 and the listed supporting resources. Prepared as a structured summary; it is not a legal opinion.